

CHAPTER 31

WIND TUNNEL PROCEDURE

31.1 SCOPE

The Wind Tunnel Procedure shall be used where required by Sections 27.1.3, 28.1.3, and 29.1.3. The Wind Tunnel Procedure shall be permitted for any building or other structure in lieu of the design procedures specified in Chapter 27 (main wind force resisting system [MWFRS] for buildings of all heights and simple diaphragm buildings with $h \leq 160$ ft (48.8 m)), Chapter 28 (MWFRS of low-rise buildings and simple diaphragm, low-rise buildings), Chapter 29 (MWFRS for all other structures), and Chapter 30 (components and cladding [C&C] for all building types and other structures).

User Note: Chapter 31 may always be used for determining wind pressures for the MWFRS and/or for C&C of any building or other structure. This method is considered to produce the most accurate wind pressures of any method specified in this standard.

31.2 TEST CONDITIONS

Wind tunnel tests, or similar tests using fluids other than air, used for the determination of design wind loads for any building or other structure, shall be conducted in accordance with this section. Tests for the determination of mean and fluctuating forces and pressures shall meet the requirements of ASCE 49.

31.3 DYNAMIC RESPONSE

Tests for the purpose of determining the dynamic response of a building or other structure shall be in accordance with Section 31.2. The structural model and associated analysis shall account for mass distribution, stiffness, and damping.

31.4 LOAD EFFECTS

31.4.1 Mean Recurrence Intervals of Load Effects. The load effect required for strength design shall be determined for the same mean recurrence interval as for the analytical method by using a rational analysis method, defined in the recognized literature, for combining the directional wind tunnel data with the directional meteorological data or probabilistic models based thereon. The load effect required for allowable stress design shall be equal to the load effect required for strength design divided by 1.6. For buildings or other structures that are sensitive to possible variations in the values of the dynamic parameters, sensitivity studies shall be required to provide a rational basis for design recommendations.

31.4.2 Limitations on Wind Speeds. The wind speeds and probabilistic estimates based thereon shall be subject to the limitations described in Section 26.5.3.

31.4.3 Wind Directionality. The directional wind climate based on recorded or simulated directional wind speed data shall be considered in determining wind loads, and the data shall be presented as part of the wind tunnel report submitted to the Authority Having Jurisdiction. The method for combining wind tunnel model data with information on wind speed and direction at the project site shall also be clearly stated in the Wind Tunnel Report. Variation in the wind direction, based upon uncertainty in the wind climate data, shall be considered when determining the wind loading, and the design wind loads shall be based on the largest values that result from this uncertainty. Consideration of uncertainty in the wind direction is not required in the determination of serviceability related wind effects.

31.4.4 Limitations on Loads. Loads for the MWFRS determined by wind tunnel testing shall be limited such that the overall principal loads in the x and y directions are not less than 80% of those that would be obtained from Part 1 of Chapter 27 or Part 1 of Chapter 28 for buildings or Chapter 29 for other structures. The overall principal load for buildings shall be based on the overturning moment for flexible buildings and the base shear for other buildings. The overall principal load for other structures shall be based on the overturning moment for flexible structures and the base shear for other structures.

Pressures for C&C determined by wind tunnel testing shall be limited to not less than 80% of those calculated for Zone 4 for walls and Zone 1 for roofs using the procedure of Chapter 30. These zones refer to those shown in Figs. 30.3-1, 30.3-2A-C, 30.3-3, 30.3-4, 30.3-5A-B, 30.3-6, 30.3-7, and 30.5-1.

The limiting values of 80% may be reduced to 50% for the MWFRS and 65% for C&C if either of the following conditions applies:

1. There were no specific influential buildings or objects within the detailed proximity model.
2. Loads and pressures from supplemental tests for all significant wind directions in which specific influential buildings or objects are replaced by the roughness representative of the adjacent roughness condition, but not rougher than Exposure B, are included in the test results.

31.5 WIND-BORNE DEBRIS

Glazing in buildings in wind-borne debris regions shall be protected in accordance with Section 26.12.3.

31.6 ROOF-MOUNTED SOLAR COLLECTORS FOR ROOF SLOPES LESS THAN 7 DEGREES

Wind loads on roof-mounted solar collectors with roof slope less than 7 degrees are permitted to be determined by wind tunnel

tests as generic loads applicable to a range of buildings, by determining load coefficients for use in the analysis equations of the Directional Procedure in Chapters 27 and 29 for MWFRS and in Part 5 of Chapter 30 for C&C. Alternatively, the generic loads are permitted to be specified with an analysis method defined in the wind tunnel test report. It is not required to include specific nearby buildings in the testing when results are to be used for multiple sites.

Wind tunnel tests shall satisfy ASCE 49, shall meet requirements specific to roof-mounted solar collectors, and shall meet the additional requirements specific to roof-mounted solar collectors, including the following variables. These requirements include accurately scaled models of solar collectors, including collector tilt angle, row-to-row spacing, aisles or gaps between collector rows compared to the main axes of the building, deflector/shroud shapes, and the geometry of the collector support structure. The tests shall include at least eight rows of collectors, where more than eight rows are applicable, mounted on the roof of representative generic buildings. The models of generic buildings shall be large enough in plan area to capture the wind flow environment over different roof zones. The test matrix shall include the range of building plan dimensions, eave height, parapet height, roof slope, and open or enclosed buildings.

Data analysis shall consider wind loads from all wind directions. Generic load coefficients shall be calculated to be consistent with coefficients in Chapters 27, 29, and 30 or shall be defined to apply to an analysis procedure specified in the test report.

The test report shall include data collection methods, data analysis, boundary layer modeling, collector and building modeling, measured wind loads and their relationship to effective wind area, conversion of data into generic coefficients, and conditions of applicability of results to different building types and collector geometry. Wind tunnel results shall not be extrapolated to geometric configurations that were not anticipated by the wind tunnel study. Interpolation between two or more tests shall be permitted. The limitations of the wind tunnel study, such as the range of collector and building geometry parameters that were tested, shall be clearly reported.

31.6.1 Wind Tunnel Test Requirements

31.6.1.1 Limitations on Wind Loads for Rooftop Solar Collectors. For photovoltaic solar collector systems that meet the limitations and geometry requirements of Fig. 29.4-7, the minimum design wind load based on a wind tunnel study shall not be less than 65% of the values resulting from 29.4-7 subject to the conditions of Section 31.4.3. The minimum design wind force based on a wind tunnel study for roof-mounted solar collector systems need not comply with the minimum net pressure of 16 lb/ft² (0.77 kN/m²) per ASCE 7-16, Section 30.2.2.

31.6.1.2 Peer Review Requirements for Wind Tunnel Tests of Roof-Mounted Solar Collectors. Wind load values lower than those indicated in Section 31.6.1.1 shall be permitted when an independent peer review of the wind tunnel test is performed in accordance with this section. The independent peer review shall be an objective, technical review by knowledgeable reviewer(s) experienced in performing wind tunnel studies on buildings and similar systems, and in properly simulated atmospheric boundary layers. The minimum qualifications for the peer reviewer shall be the following:

- The peer reviewer shall be independent from the wind tunnel laboratory that performed the tests and report and shall bear no conflict of interest.
- The peer reviewer shall have technical expertise in the application of wind tunnel studies on buildings similar to that being reviewed.
- The peer reviewer shall have experience in performing or evaluating boundary layer wind tunnel studies and shall be familiar with the technical issues and regulations governing the Wind Tunnel Procedure in ASCE 49 as it is applied to systems similar to solar photovoltaic collectors that use generalized wind tunnel data for design.

The peer reviewer shall review the wind tunnel report, including but not limited to data collection methods, data analysis, boundary layer modeling, collector and building modeling, resulting wind loads and their relationship to effective wind area, conversion of data into GC_{rn} values, and conditions of applicability of results to different building types, collector geometry, and other relevant issues identified by the reviewer.

The peer reviewer shall submit a written report to the Authority Having Jurisdiction and the client. The report shall include, at a minimum, statements regarding the following: scope of peer review with limitations defined; the status of the wind tunnel study at time of review; conformance of the wind tunnel study with the requirements of ASCE 49 and Section 31.6.1; conclusions of the reviewer identifying areas that need further review, investigation, and/or clarification; recommendations; and whether, in the reviewer's opinion, the wind loads derived from the wind tunnel study are in conformance with ASCE 7-16 for the intended use(s).

31.7 CONSENSUS STANDARDS AND OTHER REFERENCED DOCUMENTS

This section lists the consensus standards and other documents that shall be considered part of this standard to the extent referenced in this chapter.

ASCE 49, *Wind Tunnel Testing for Buildings and Other Structures*, American Society of Civil Engineers, 2012.

Cited in: Sections 31.2, 31.6, 31.6.1.2, C31, C31.4.2, C31.6.1